WEEK 1 · DAY 3 · OVERVIEW

Market Calendars and Cleaning



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ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Before anyone takes a change, a window or a join on our curve, I need to know which days the market had. Put the weekdays our Treasury file skips in a function, and give Excel the same list. Everyone downloads a holiday calendar: tell me whether it agrees with the file. Check that FRED's dots fall on those same days. Then put series side by side without inventing a day, show me what forward fill does to a change, and find any value that stopped updating. Three tested functions, committed under a message we can search for.

ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

Your Route Through Day 3

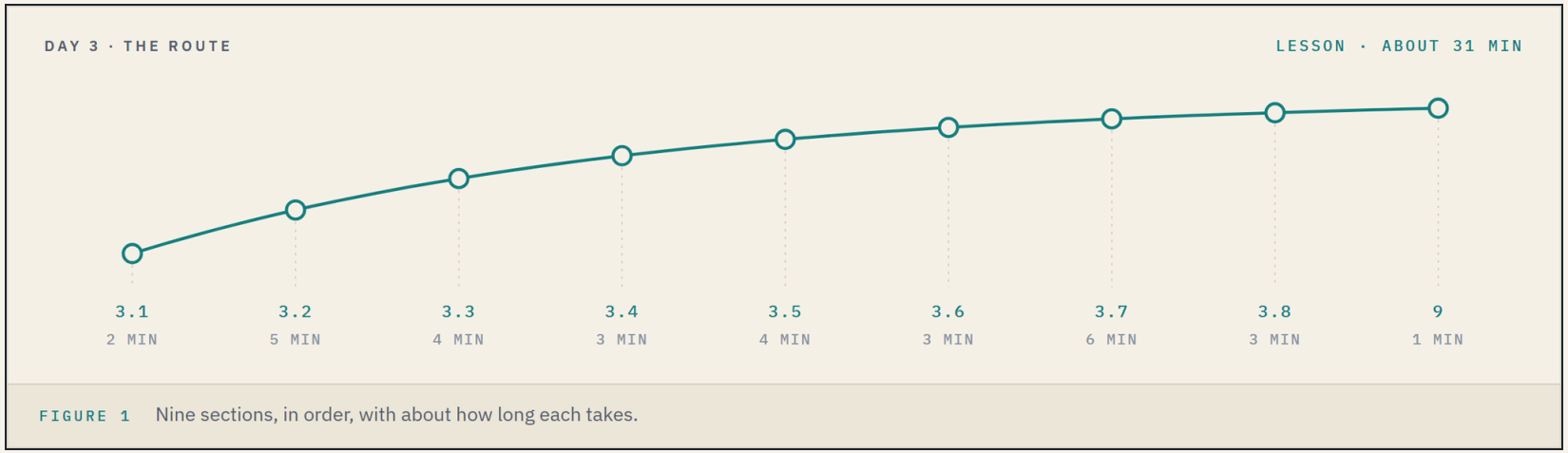


FIGURE 1 Nine sections, in order, with about how long each takes.

	SECTION	ABOUT
3.1	Three Kinds of Day	2 min
3.2	The File as Its Own Calendar: `missing_weekdays`	5 min
3.3	A Holiday Calendar Is Not the Market Calendar	4 min
3.4	FRED's Dots Are the Same Days	3 min
3.5	Joining Two Series: `align`	4 min

	SECTION	ABOUT
3.6	Forward Fill, for Display Only	3 min
3.7	Stale Values: `stale_runs`	6 min
3.8	Fix the Message Before Anything Builds on It	3 min
9	Bring It Home	1 min

Time. About 31 minutes for the lesson, then about 12 for the four exercises.

3.1 Three Kinds of Day

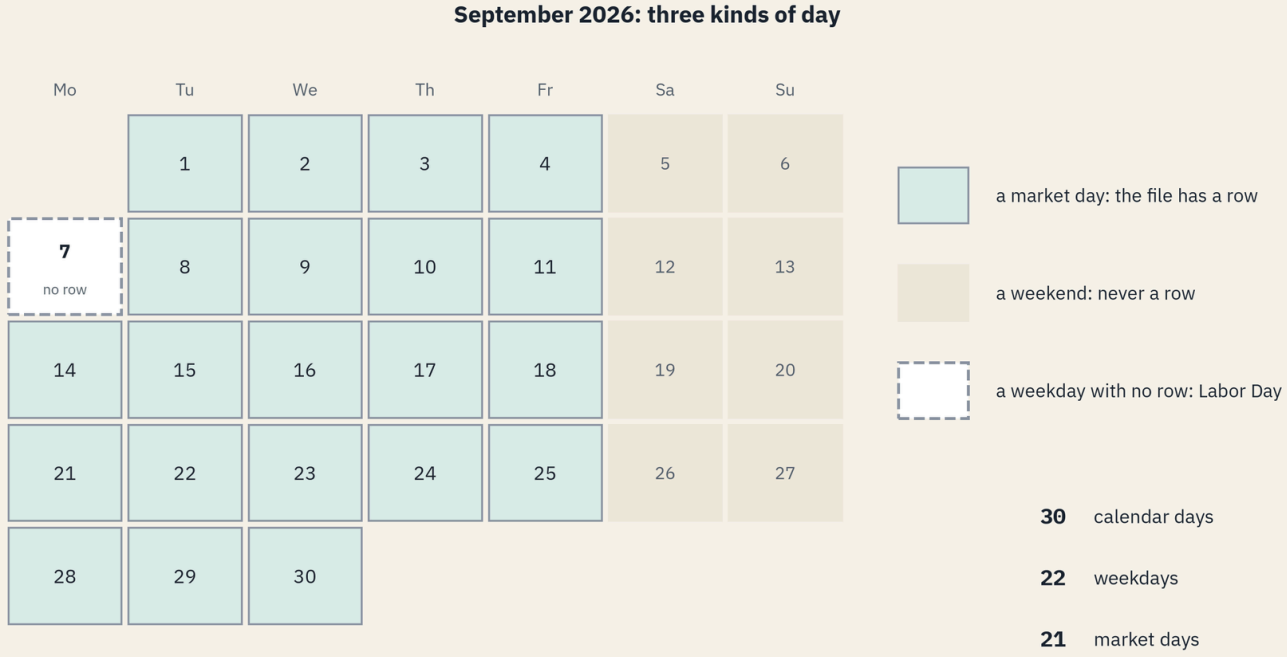


FIGURE 2 September 2026 has 30 calendar days, 22 weekdays, and 21 market days: Labor Day has no row.

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Calendar days	`=A2941-A2+1`	4,292
Weekdays	`=NETWORKDAYS(A2,A2941)`	3,066
Market days	`=COUNT(A2:A2941)`	2,940

Calendar days, weekdays, market days: every calculation in this course runs on the third.

3.2 The File as Its Own Calendar: missing_weekdays

THE SEVEN OF 2026

```
for day in missing_2026:
    print(day.date(), day.strftime('%A'))
```

```
2026-01-01 Thursday
2026-01-19 Monday
2026-02-16 Monday
2026-05-25 Monday
2026-06-19 Friday
2026-07-03 Friday
2026-09-07 Monday
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Weekdays of 2026, to 2 October	`=NETWORKDAYS(DATE(2026,1,1),DATE(2026,10,2))`	197
Less the list	`=NETWORKDAYS(DATE(2026,1,1),DATE(2026,10,2),H2:H8)`	190
September, less the list	`=NETWORKDAYS(DATE(2026,9,1),DATE(2026,9,30),H2:H8)`	21

`missing_weekdays` **makes the file its own calendar: 126 dates, and as Excel's holiday list the seven of 2026 give the file's 190 rows.**

3.3 A Holiday Calendar Is Not the Market Calendar

PANDAS' FEDERAL HOLIDAYS

```
from pandas.tseries.holiday import GoodFriday, USFederalHolidayCalendar
federal = USFederalHolidayCalendar().holidays(first, last, return_name=True)
print(len(federal), 'federal holidays, all on weekdays:', (federal.index.dayofweek < 5).all())
```

```
121 federal holidays, all on weekdays: True
```

ITS GOOD FRIDAY RULE

```
good_fridays = GoodFriday.dates(first, last)
```

> **The course gives no reason for any of these days: the file is the record.**

3.4 FRED's Dots Are the Same Days

THE SAMPLE'S DOT

```
sample = md.parse_fred_observations(payload, 'DGS10')
dots = sample.index[sample.isna()]
print('Dots:', list(dots.strftime('%Y-%m-%d')),
      '| each a missing weekday:', dots.isin(missing).all())
```

```
Dots: ['2026-09-07'] | each a missing weekday: True
```

THE NUMBERS

```
Same dates: True | largest gap: 0.0 percent
```

THE LIVE CELL, ITS WORKING LINES

```
    live_span = live_dgs10.loc[first:live_end]
    live_dots = live_span.index[live_span.isna()]
```

WITH YOUR KEY

With your key, it prints FRED's rows since 2015, its dots, every one a weekday the file skips, and a largest gap of 0.

FRED's dots and the file's absent rows are the same days: drop the dots and the two sources share one calendar.

3.5 Joining Two Series: align

THE FUNCTION

```
def align(series: dict[str, pd.Series], how: str = "inner") -> pd.DataFrame:
    frame = pd.concat(series, axis=1, join=how)
```

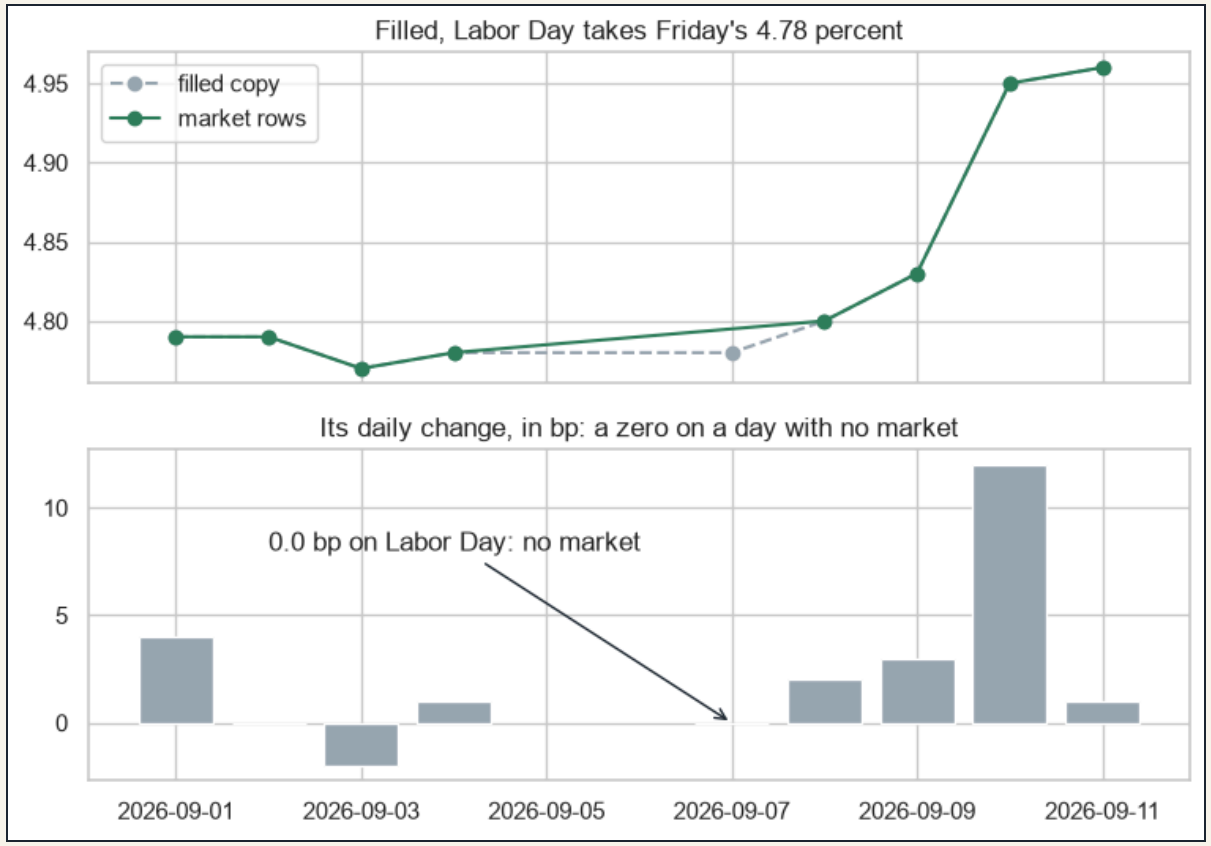
WHERE OUTER EARNS ITS PLACE

```
four_month = treasury['4 Mo'].dropna()
inner = md.align({'4 Mo': four_month, 'ig_spread_bp': spreads['ig_spread_bp']})
outer = md.align({'4 Mo': four_month, 'ig_spread_bp': spreads['ig_spread_bp']}, how='outer')
print(len(inner), 'rows inner;', len(outer), 'outer, with', outer['4 Mo'].isna().sum(), 'NaN')
```

```
989 rows inner; 2940 outer, with 1951 NaN
```

Inner drops almost eight years of spreads; outer keeps them and leaves the bill's cells empty, as they were.

3.6 Forward Fill, for Display Only



EXCEL SIDE BY SIDE

JOB	EXCEL
Fill forward	`=IF(T3="",U2,T3)`
Labor Day's change	`=(U3-U2)*100`
Tuesday's change	`=(U4-U3)*100`

Forward fill invents a zero change on a day with no market: fill a displayed copy only, never before a change, a window, or a join that feeds a number.

3.7 Stale Values: stale_runs

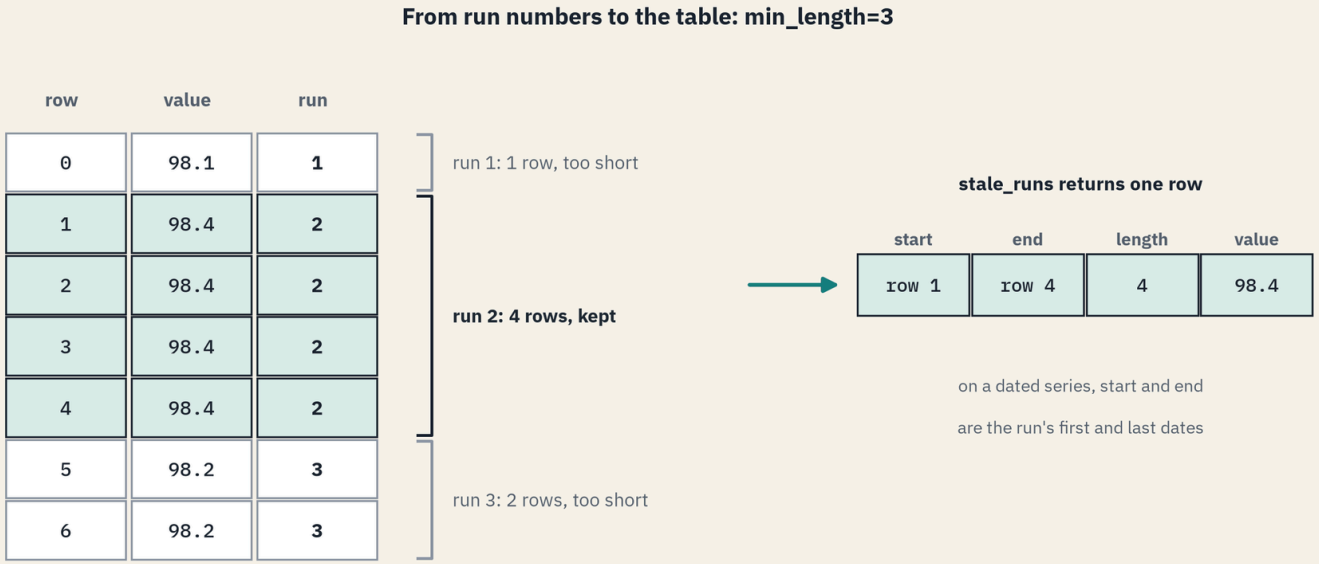


FIGURE 5 Seven values grouped by run: only the run of four at 98.4 passes min_length 3 and becomes a table row.

RUNS OF THREE IN THE SPREADS

```
4 IG runs of 3 or more; 0 HY
       start        end  length  value
0 2015-02-13 2015-02-18       3   84.7
1 2021-12-08 2021-12-10       3   81.3
2 2024-02-14 2024-02-16       3  107.2
3 2025-11-21 2025-11-25       3   92.2
```

EXCEL SIDE BY SIDE

JOB	EXCEL
Runs of 3 or more	`=COUNTIF(D4:D2941,TRUE)`
Where	the rows showing TRUE

A run is a reason to look, not a verdict: no run of five or more in the spreads, and a two-decimal yield sat still for eight market days.

3.8 Fix the Message Before Anything Builds on It

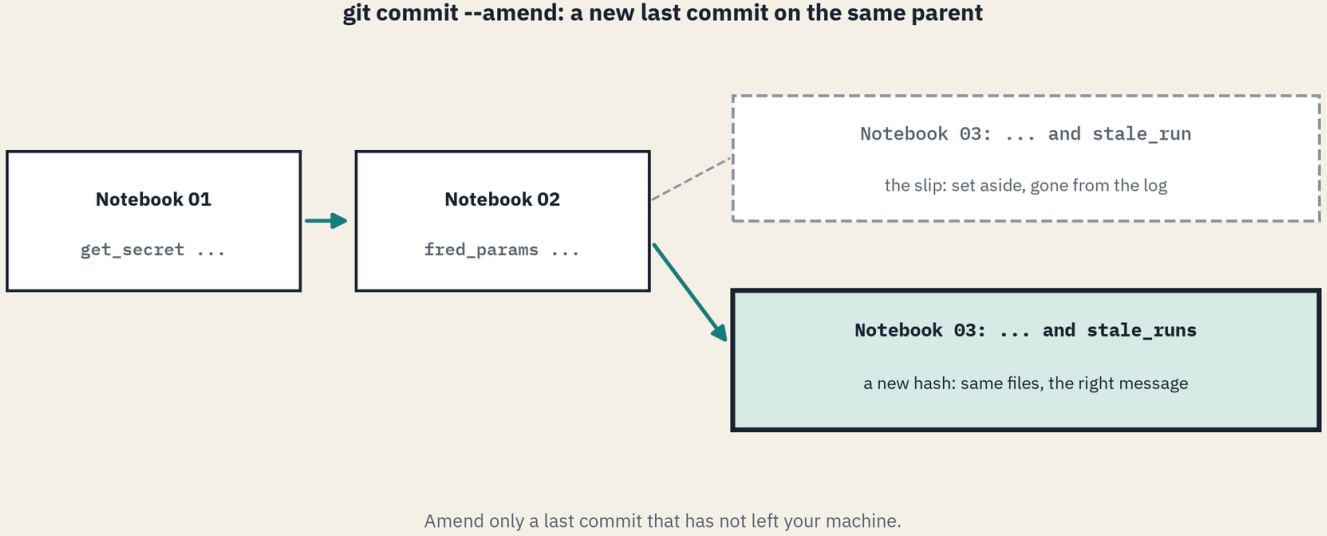


FIGURE 6 Amending replaces the slipped commit with a new hash on the same parent, with the same files.

WHAT AMEND DOES

`git commit --amend` swaps the last commit for a new one with the same files; `-m` gives it the right message.

WHEN TO AMEND

Amend only a last commit that has not left your machine: once pushed, fix it with a new commit.

`--amend` **replaces the last commit: the same files, the right message, and the slip gone from the log.**

Copy your two files home and commit.

Bring It Home

IN A POWERSHELL TERMINAL

```
# the work folder: use the name your setup cell printed
$work = "$env:TEMP\pfi_course3_03_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\marketdata.py", "$work\test_marketdata.py" .
python -m pytest -q test_marketdata.py
git status --short
git diff --stat
git add marketdata.py test_marketdata.py
git commit -m "Notebook 03: missing_weekdays, align, and stale_runs"
```

pytest shows 13 passed, and the DGS10 sample the tests read is there since notebook 02.

You Can Now

- Count calendar days, weekdays and market days, and say which one a calculation runs on.
- Find the weekdays a file skips with **`missing_weekdays`**, and give Excel the same list.
- Show where pandas' federal holiday calendar disagrees with the file, both ways.
- Check that FRED's dots fall on the file's missing weekdays.
- Join series on their dates with **`align`**, inner or outer, filling nothing.
- Say what forward fill does to a change, and keep it for display.
- Find values that stopped moving with **`stale_runs`**, and fix a commit message with **`--amend`**.

NEXT

WEEK 1 · DAY 4

Resampling and the As-Of Value

One number for a week or a month: which observation stands for it, the date it really comes from, a week with no Friday, what a calendar grid does to a holiday, and the value as of any date, tied to Excel.

